VOYIS VSLAM POWERED BY EIVA NAVISUITE

QUICK START GUIDE

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1 Introduction

Welcome to the quick start guide for Voyis VSLAM Powered by EIVA NaviSuite. This document will help you get started with our licenced software package which includes two key applications:

- **Discovery Powered by NaviSuite:** This is the graphical user interface (GUI) program that you will interact with.
- **NaviSuite VSLAM:** This background service processes data from the Voyis Discovery Stereo camera to perform visual simultaneous localisation and mapping (VSLAM).

By following this guide, you'll learn the basics of using these applications effectively.

Note: This software is not intended to run with the Voyis Discovery GUI. Running Voyis VSLAM Powered by EIVA NaviSuite while running Voyis Discovery GUI will cause unstable and unintended behaviour.

2 Discovery Powered by NaviSuite – Getting started

First, install the Voyis VSLAM Powered by EIVA NaviSuite software and activate the licence as described in the Voyis VSLAM Powered by EIVA NaviSuite Installation guide.

Next, open Discovery Powered by NaviSuite and create a new project:

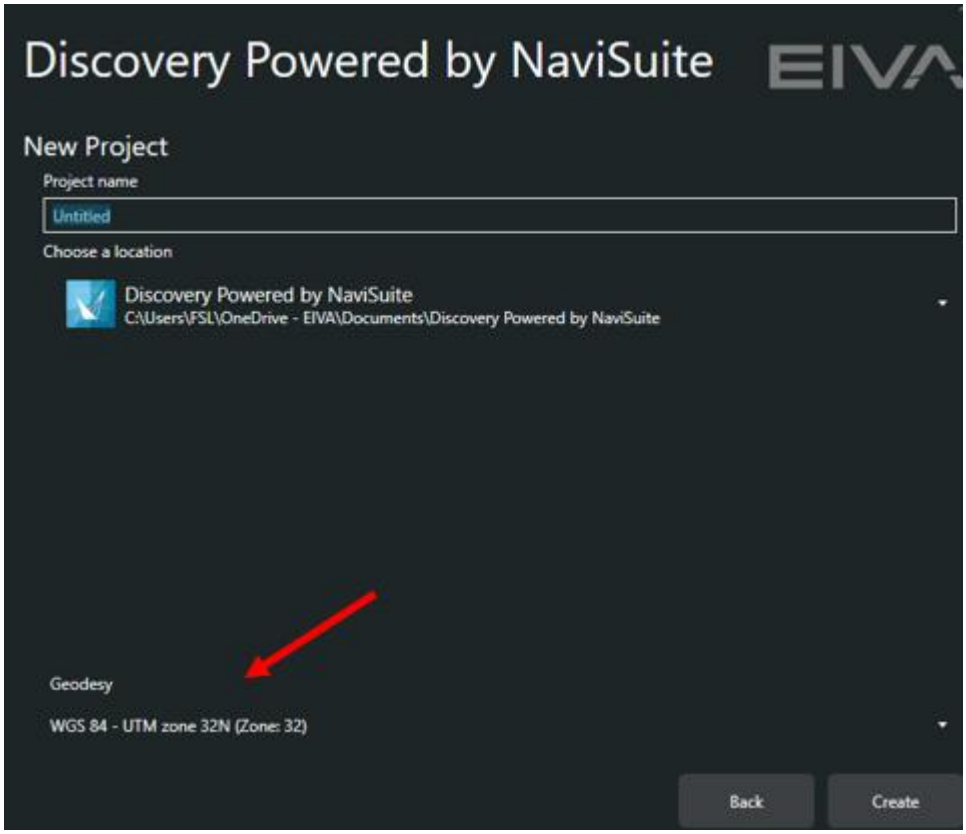

Figure 1 Discovery Powered by NaviSuite software – Create a new project

The geodesy for the project is chosen when starting a new project. In the bottom left corner, when clicked on, the separate window for choosing Geodesy will open. If you wish to change Geodesy once the project has been created, you can do so from the Settings menu.

You can read more about geodesy on our knowledge base.

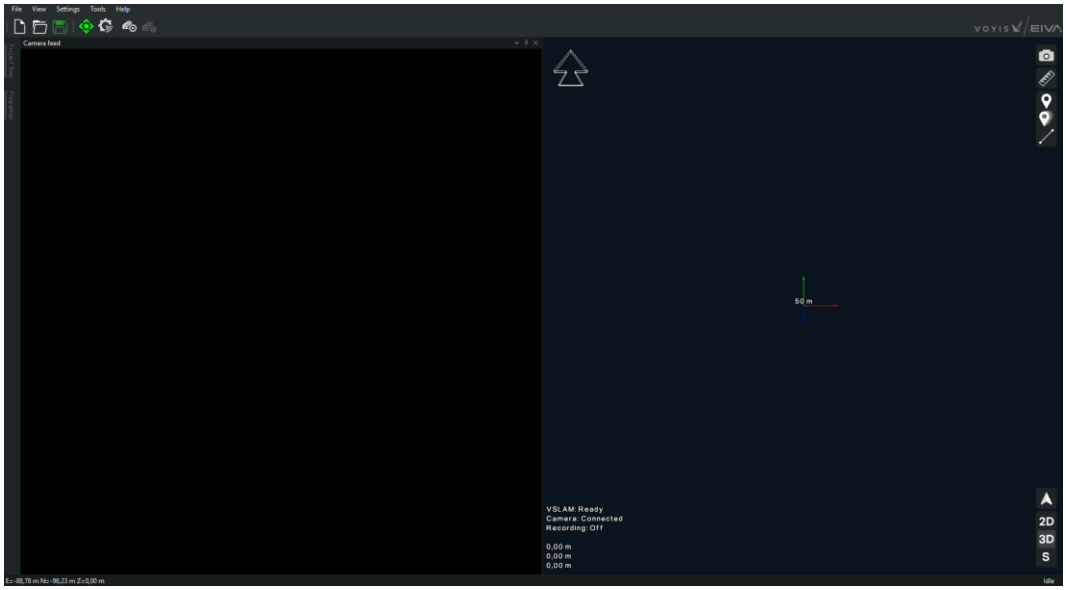

Figure 2 Discovery Powered by NaviSuite – A new project showing the **Camera feed view** (left) and **Map view** (right)

If the camera is connected to the same network, it is automatically connected and shows the status **Camera: Connected** in the bottom left of the **Map view**.

Figure 3 Discovery Powered by NaviSuite – Camera: Connected

All the files generated by the software will be automatically saved inside the project folder structure. A configurable **Auto Save Project** option is also available from **File** > **Auto Save Project**.

Project settings are saved alongside project files. When you open an existing project, these will be imported automatically.

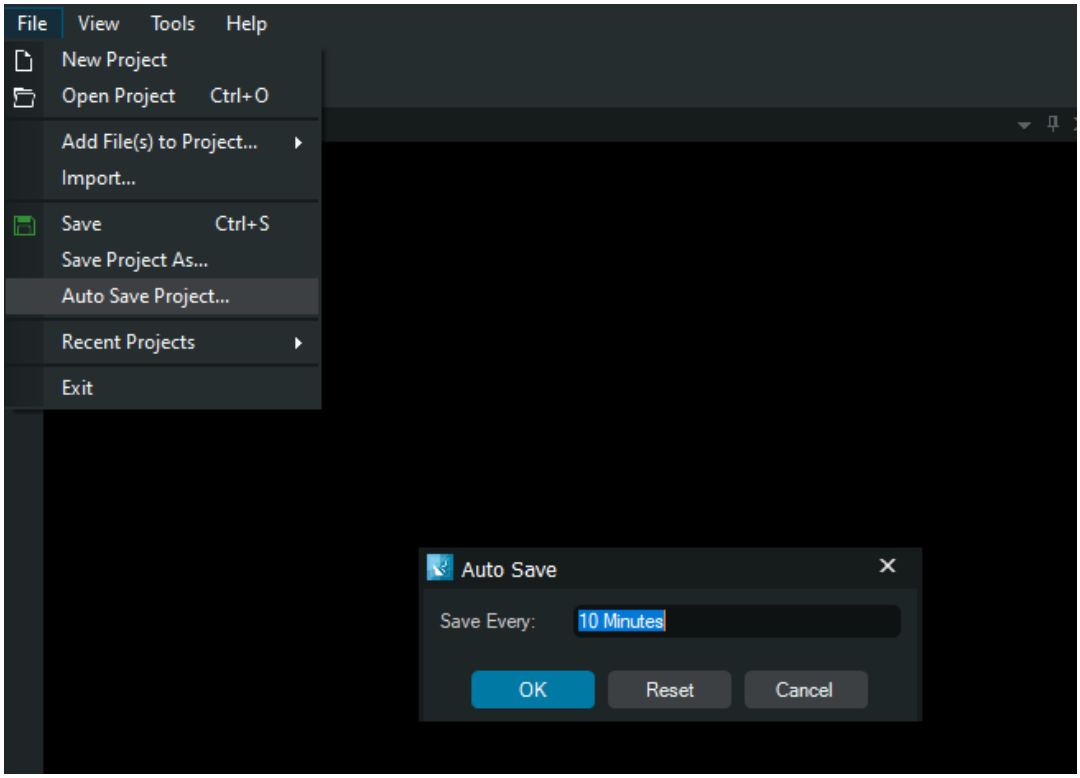

Figure 4 Discovery Powered by NaviSuite – Auto Save Project option

2.1 Operating recommendations

For NaviSuite VSLAM to perform as expected, the following operating conditions should be met:

- Ethernet bandwidth must be at least 500 Mbps
- The camera should be configured to stream stills at 10 fps
- The camera needs to be oriented towards a structure to function properly. The camera must stay consistently between 0.5 m and 1 m (optimally) from the target object

- The vehicle should be moving at a maximum speed of 0.2 m/s
- The vehicle needs to move in a stable pattern (i.e. the attitude needs to be consistent)

Conditions where the software can present malfunctioning behaviour:

- Absence of overlapping images
- Absence of features in the object
- Insufficient and over saturated lighting conditions
- Poor visibility/turbidity of the water
- Movement at speeds exceeding 0.2 m/s

2.2 Discovery Powered by NaviSuite toolbar

The NaviSuite VSLAM service running behind Discovery Powered by NaviSuite is controlled from this section of the Discovery Powered by NaviSuite toolbar.

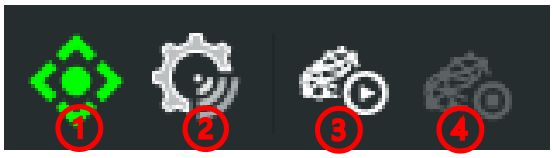

Figure 5 Discovery Powered by NaviSuite – NaviSuite VSLAM toolbar

1. **Camera Control**
2. **Sensor Settings**
3. **Start NaviSuite VSLAM**
4. **Stop NaviSuite VSLAM**

2.2.1 Camera Control

Before starting the NaviSuite VSLAM service, it is possible to adjust using the **Camera Control** settings.

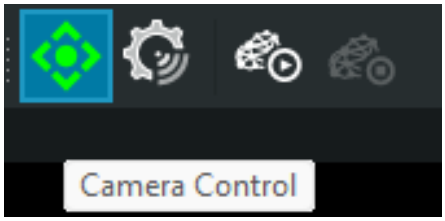

Figure 6 Discovery Powered by NaviSuite – Camera Control

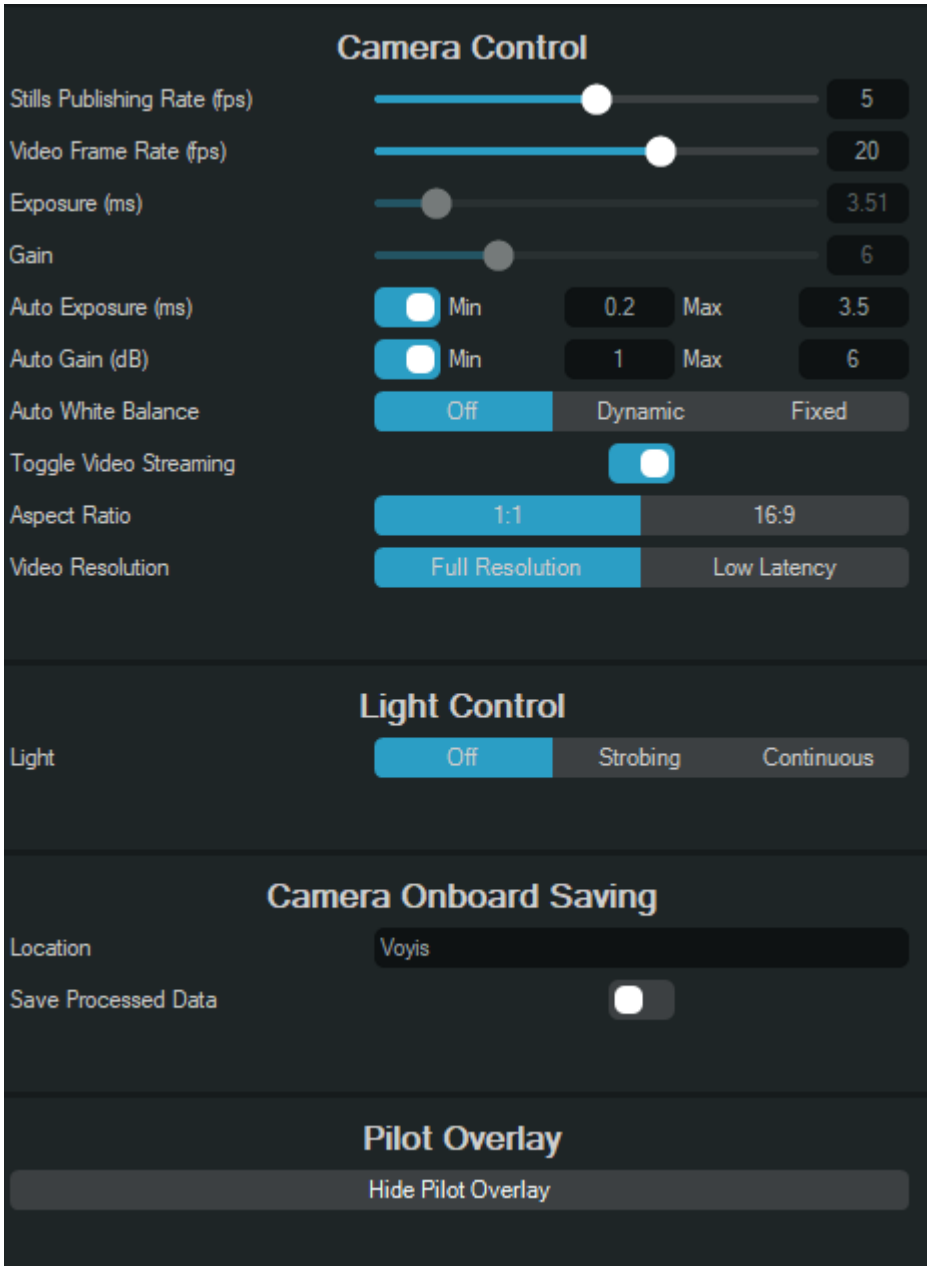

Figure 7 Discovery Powered by NaviSuite – Voyis Discovery Camera Control view

- **Stills Publishing Rate (fps)** – Enables data saving on the topside computer. The default value and recommended minimum is 5, the maximum allowed is 10.
    - **Note**: Reducing this value below 4 fps will cause issues with the VSLAM output.
    - The saving rate of processed images will be impacted by the complexity and bandwidth of the network connection the control computer has to Discovery Powered by NaviSuite. It is recommended that you conduct

reliability testing on your network before a survey. If these rates cannot be achieved using Data Publishing Rate, it is recommended that Camera Onboard Data Saving be used to save processed images instead.
    - **Note**: RAW images are always saved without any image processing or undistortion. The ability to toggle image undistortion has been removed from the GUI to ensure users will always collect calibrated images during data collection. Users may still collect un-calibrated images via the SDK / API interface.
- **Video Frame Rate (fps)** – Video frame rate in frames per second.
    - Recommended: 20 fps
    - Valid range: 1 fps to 20 fps
    - Set Video frame rate should be set at 1x or 2x the Stills Publishing Rate
- **Exposure (ms)** – Default is 2 ms, but it is possible to change manually. Higher exposure results in increased image brightness as more light is received by the sensor.
    - Recommended (per Light Control Mode):
    - **Off** – 0.2 ms to 30 ms
    - **Strobing** – 0.2 ms to 3.5 ms (max strobe duration is 3.5 ms)
    - **Continuous** – 0.2 ms to 30 ms
- **Valid range:** 0.2 ms to 30 ms
    - **Note**: The user should minimise exposure time, while still ensuring a sufficiently illuminated video. A lower exposure will reduce image blur caused by vehicle movement, which improves the results of 3D modelling.
- **Gain (db)** – Set by default. This controls the camera sensor sensitivity.
    - Recommended: 1 dB to 6 dB (>6 dB may cause image graininess)
    - Valid range: 1 dB to 20 dB
    - **Note**: Gain can be used to increase image brightness without the need to increase exposure that may increase image blur. This is valuable in Strobing mode (max exposure of 3.5 ms), or when using ambient sunlight (Light status: Off).
- **Auto Exposure (ms**) – Automatically optimises camera exposure time to achieve a reasonable image brightness.
    - Recommended: On
    - The recommended Min and Max depend on Light status in the Light Control option:
    - **Off** – 0.2 ms to 30 ms.
    - **Strobing** – 0.2 ms to 3.5 ms.
    - **Continuous** – 0. ms to 20 ms. If capturing images for 3D modelling, keep the max exposure below 5 ms to reduce blur.
- **Auto Gain (dB)** – Automatically optimises camera gain to achieve a reasonable image brightness.
    - Recommended: On
    - Valid range: 1 dB (min) to 20 dB (max)
    - Note: Exceeding 6 dB can increase image graininess/noise
- **Auto White Balance** – Automatically corrects image colour in real time by adjusting colours to balance blue, green, and red channels.
    - Recommended: Fixed
    - **Off** – No white balance
    - **Dynamic** – Continually adjusted white balance

    - **Fixed** – When clicked, a one-time white balance adjustment will be captured that will be applied to images. This adjustment will be used unless it is clicked again or another white balance mode is selected.
- **Toggle Video Streaming** – Turns off the video stream. Useful when RTSP video stream is being utilised in a separate viewer, and this GUI is being used only for selecting settings. Bandwidth is doubled if this is not disabled when RTSP is being used.
- **Aspect Ratio** – Camera stream ratio. It will impact video and collected stills. 1:1 and 16:9 aspect ratios can be chosen.
- **Video Resolution** – Video stream and video saving resolution, lower video resolution will reduce latency, as outlined below:
    - Recommended: Low Latency (1080p)
    - **Low Latency** (1080p) – Prioritises minimising delay for a smoother viewing experience in 1080p resolution
    - Approx. average latency: 141 ms; 130 ms in full-screen mode
    - Approx. minimum latency: 110 ms; 99 ms in full-screen mode
    - **Note**: Running the GUI in full screen mode reduces latency further
    - **Full Resolution** (2816p) – Ideal for saving the video stream in ultra-high resolution, preserving excellent visual quality and detail
    - Approx. average latency: 186 ms; 180 ms in full-screen mode
    - Approx. minimum latency: 160 ms; 160ms in full-screen mode
- **Light**: This option controls the status of the lights on the Voyis Discovery Stereo camera.
    - Recommended: Strobing
    - **Off** – No light. LEDs do not turn on for image capture. Useful when utilising ambient sunlight as light source. Requires a higher exposure time and gain.
    - **Strobing** – LEDs flash (strobe) synchronised to camera exposures. Enables maximum light output for maximum operating range, while using short exposures of up to 3.5 ms for crisp images.
    - **Continuous** – LEDs are continuously on. Reduces light output to 20% of the maximum but enables simpler use with other camera systems.
- **Location** – Click in box to setup a data folder on the product's onboard hard-drive.
- **Save Processed Data** – Saves processed images (.jpg format, 8-bit) to onboard product storage. Processed images lighting and colour correction.
- **Pilot Overlay** – refer to section 4.3.1

2.2.2 Sensor Settings

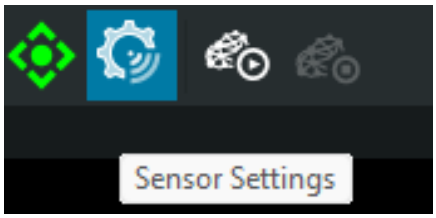

Figure 8 Discovery Powered by NaviSuite – Sensor Settings

Under **Sensor Settings**, the user can setup INS tagging and logging of INS data on the camera. When connected to an INS, the images logged by the Discovery Mono or Stereo

systems will be geotagged in real time. This will improve photogrammetry accuracy and allow for scaled models to be generated from the Discovery Stereo and Mono Camera. INS will provide latitude, longitude and altitude and each image will have that metadata populated through EXIF.

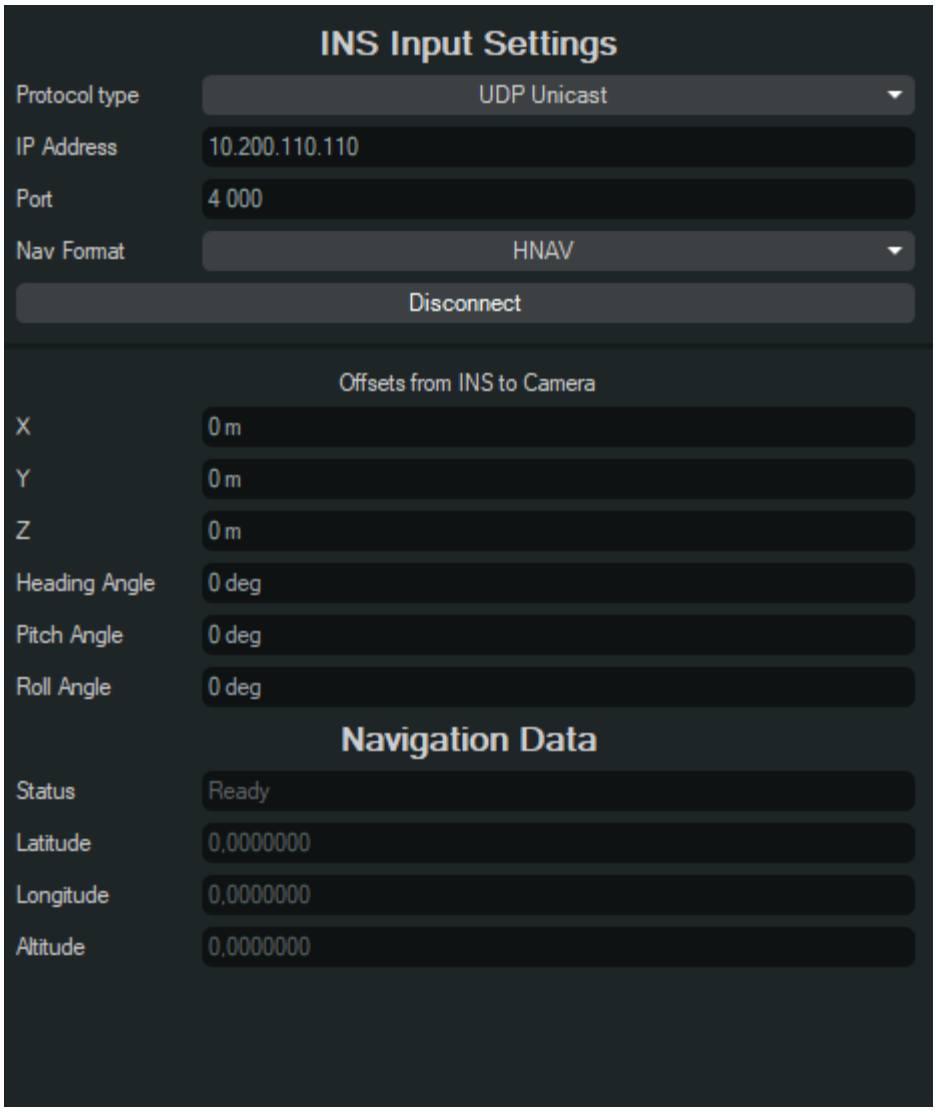

Figure 9 Discovery Powered by NaviSuite – Sensor Settings window

The fields for the Sensor Settings window are covered in the section below.

2.2.2.1 INS Input Settings

Type: UDP, TCP Server or TCP Client

Connect: Once setup is complete this will connect to the INS data stream. If successful, the ROV icon will turn green. If there is an error a message will appear, the **Status** in the display box will change and the ROV icon will turn red.

It is important to ensure that the INS is aligned before pressing the connect button. INS devices should automatically align themselves on startup, but it can take a few minutes.

The status messages are in the table below.

Status	Summary
Ready	Ready to connect or receive valid data
Receiving Data	Receiving parse-able data
Disabled	Manually disconnected
Warning: No Data Available	This state can occur if INS is setup but not streaming data and the image recording has started.
Warning: Data Invalid	Data received does not match navigation format
Warning: Data Stopped	Stopped receiving data stream for 5 seconds
Warning: Data Not Sync'd	INS and image timestamps do not match within 5 seconds
Error: Failing	Internal setup error. Try to restart the camera or contact Voyis support
Error: Invalid Settings	Incorrect INS Input Settings
Error: Timeout	Camera and INS connection is lost due to timeout

Table 1 Status messages

Nav Format: Select from a list of allowed navigation formats. The INS needs to be configured to output data using the same format.

The following formats are accepted:

- HNAV
- PSONNAV
- LNAVUTC
- Hydroid
- Delphins
- Rovins
- Navlab
- IxBlueSTD
- PhinsSTD
- NMEA (Required: GGA; Optional: GLL, HDT, PRDID)

IP Address: The IP address of the connection

Port: The port number of the connection

2.2.2.2 Offsets from INS to Camera

Figure 10 Discovery Powered by NaviSuite – Offsets from INS to Camera

Measurements are done from INS (CRP) to camera origin. The exact location of the camera origin relative to product mounting locations is provided in the product Integration Drawing. Please contact Voyis technical support at support@voyis.com if technical drawings are needed.

The offset is defined as the X, Y, Z distance in metres, from the primary position/navigation sensor to the sensor, where X is positive starboard, Y is positive front and Z is positive up, as shown in figure below.

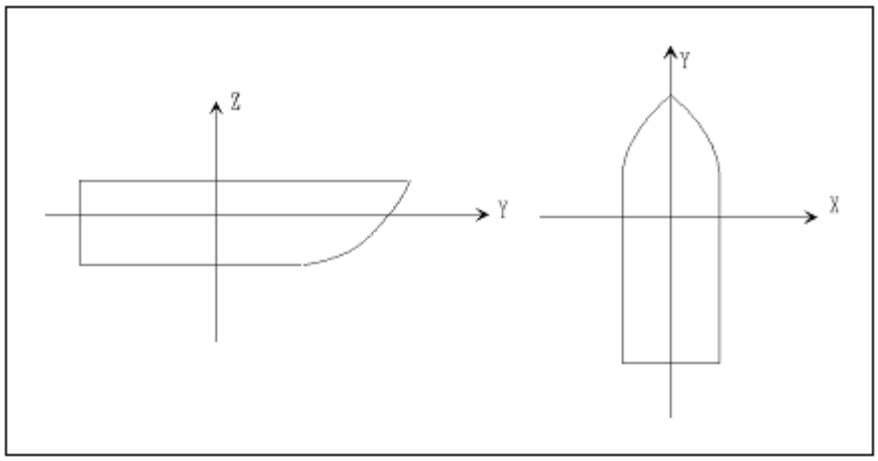

Figure 11 Discovery Powered by NaviSuite – Offsets convention

The mount angle defines the exact alignment of the sensor.

The mounting is specified by roll angle, pitch angle and heading angle; the roll angle is measured positive starboard down, the pitch angle positive bow up and the heading angle zero front and positive clockwise as shown in the figure below:

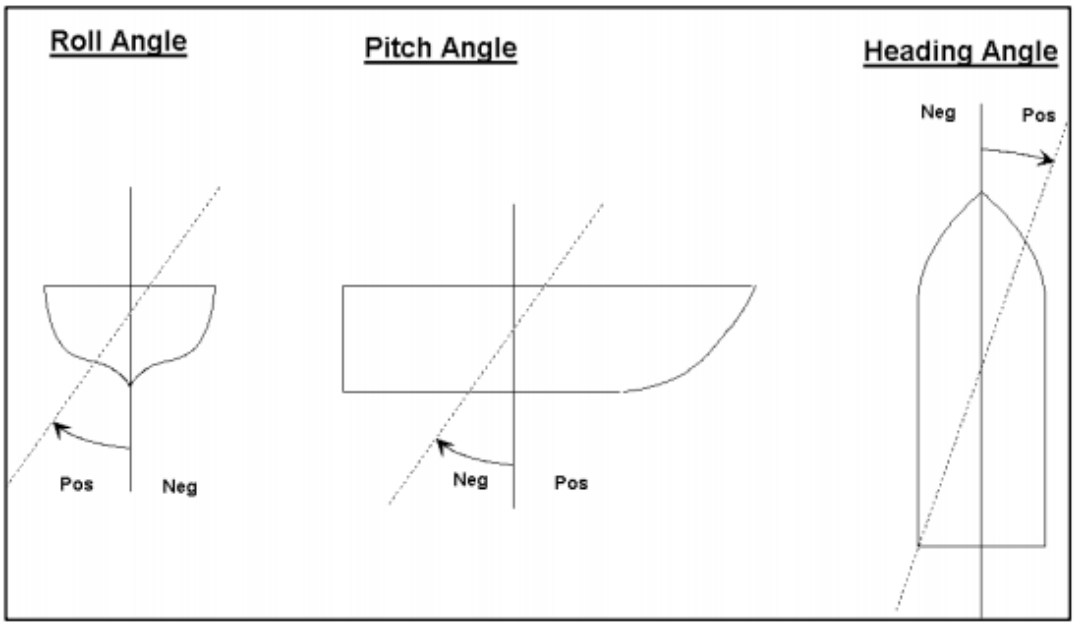

Figure 12 Discovery Powered by NaviSuite – Mounting conventions

2.2.3 Starting and stopping NaviSuite VSLAM

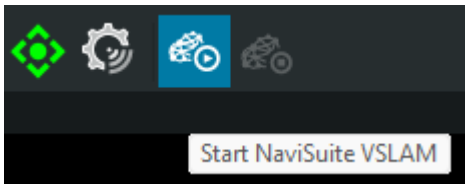

Figure 13 Discovery Powered by NaviSuite – Start NaviSuite VSLAM

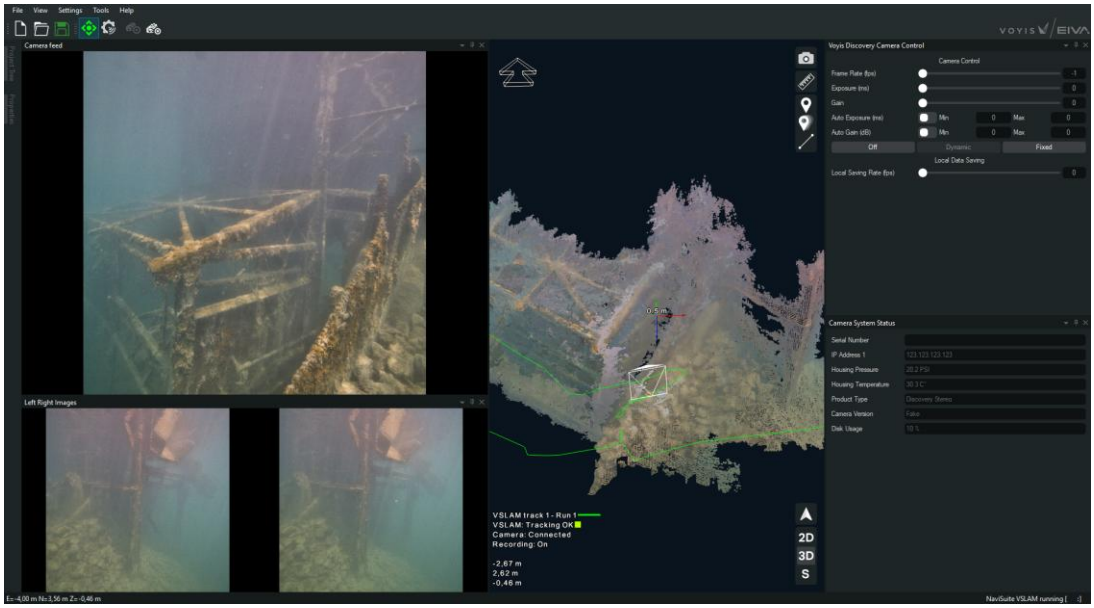

Figure 14 Discovery Powered by NaviSuite with video and point cloud

When starting NaviSuite VSLAM, Discovery Powered by NaviSuite starts building a track and a point cloud in the **Map view**.

The track and the point cloud are automatically added to the **Project Tree**.

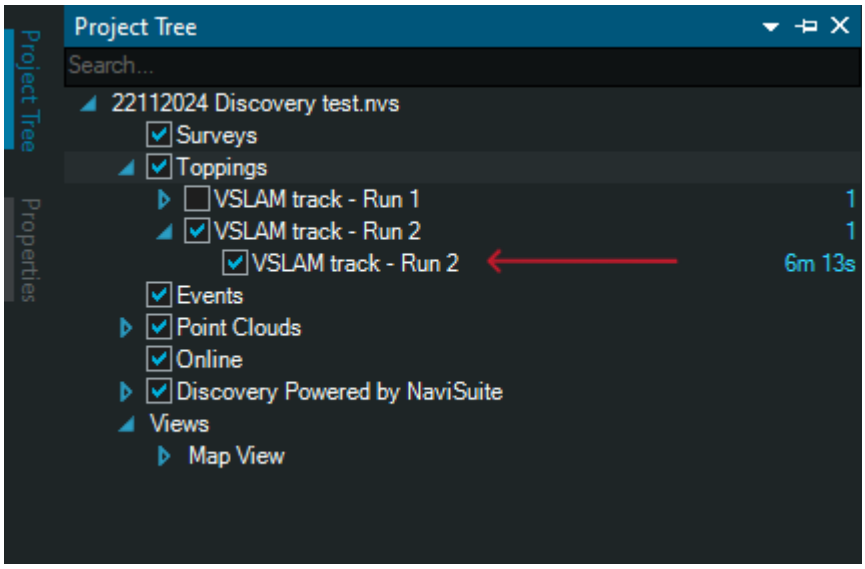

Figure 15 Discovery Powered by NaviSuite – Project Tree with track

The track is automatically saved inside the project folder when the project's **Toppings** folder is saved.

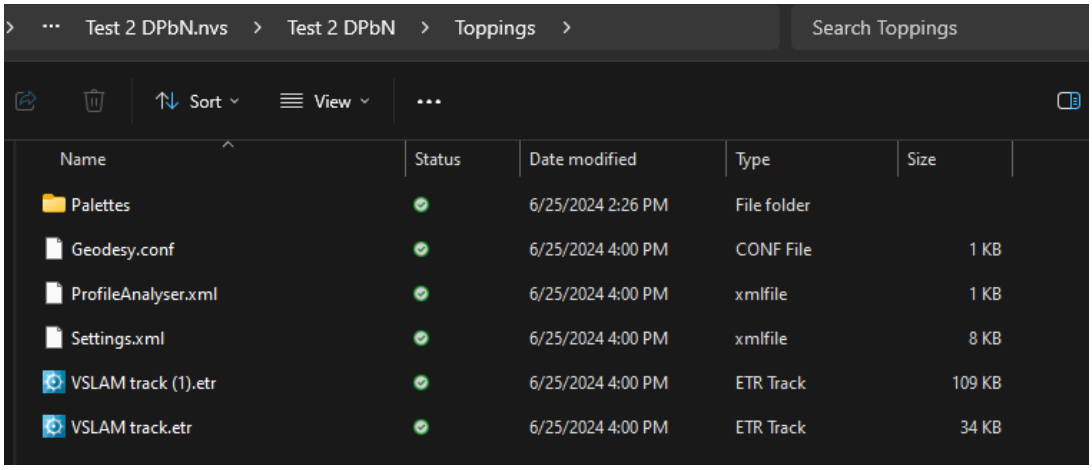

Figure 16 Toppings folder

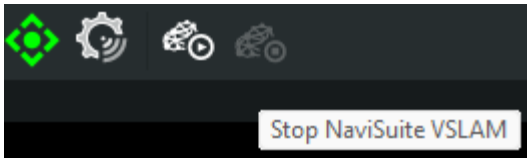

Figure 17 Discovery Powered by NaviSuite – Stop NaviSuite VSLAM

When you have finished logging, use the **Stop NaviSuite VSLAM** button.

Every time you start NaviSuite VSLAM, a new track is started from position *0,0,0* and a new point cloud is started.

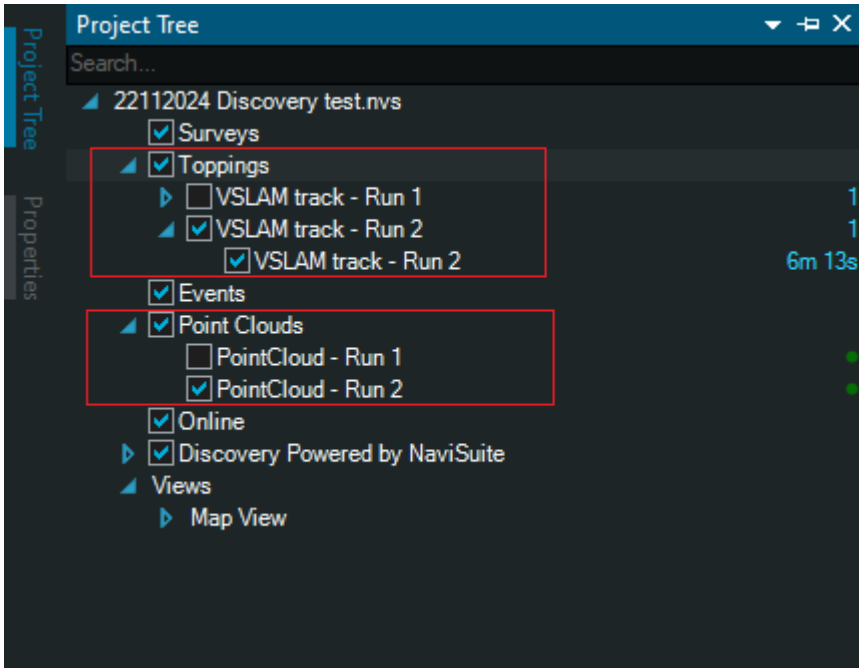

Figure 18 Discovery Powered by NaviSuite – Point clouds and VSLAM tracks

To start over, remove previous track and point clouds by right clicking and selecting **Remove,** then start NaviSuite VSLAM again.

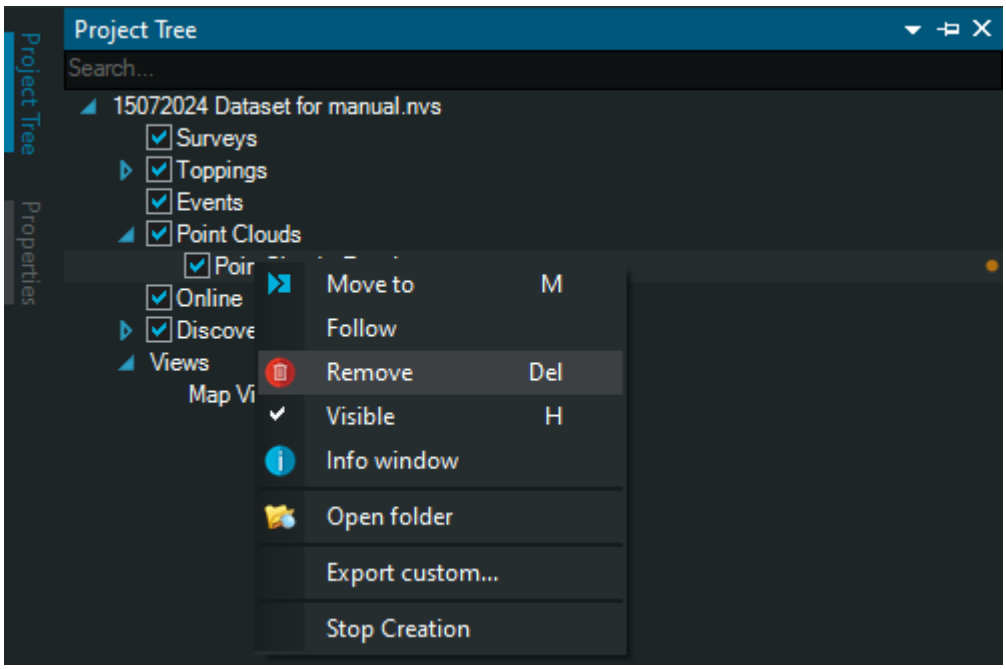

Figure 19 Discovery Powered by NaviSuite – Remove point clouds

2.2.4 Record images

Figure 20 Discovery Powered by NaviSuite – Recording: On

Discovery Powered by NaviSuite starts recording automatically when starting the NaviSuite VSLAM process. The status is shown under the **Map view** where it will show *Recording: ON* while the NaviSuite VSLAM process is running.

The images are saved to the folder *NaviSuite vSLAMCache* into a folder named *Run_xxxx* where xxxx represents the date and time in yyyyMMdd and time in 24 hr format.

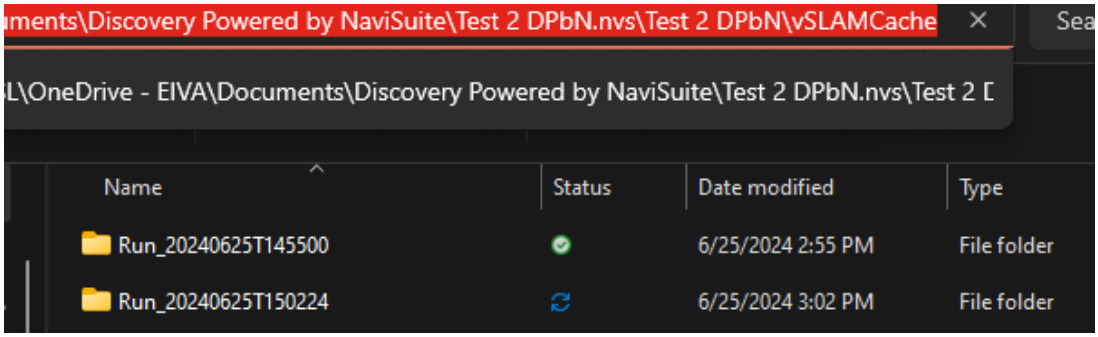

Figure 21 NaviSuite VSLAMCache folder with the image folder here called *Run_20240604T09434*

Each folder can be used to play back data. The processed subfolder has the pictures from the left and right cameras and the .YAML file.

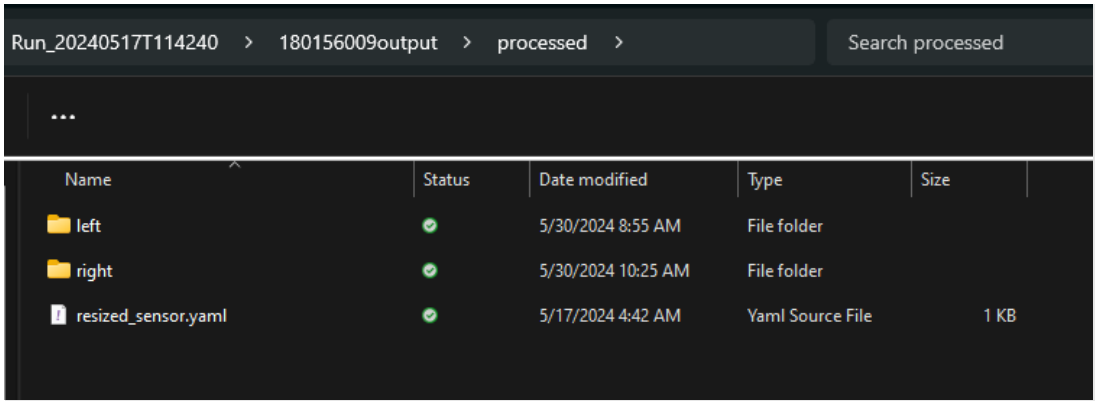

Figure 22 NaviSuite VSLAM folder structure

2.2.5 Saving INS and IMU Data

Once INS setup has been configured in Discovery Powered by NaviSuite, the INS and IMU data collected alongside the Voyis Discovery camera, will be saved in the following locations.

Note: *project path* is a placeholder to represent where your project is saved.

2.2.5.1 IMU calibration data save location

project_path\vSLAMCache\run\ReplayCamera0output\processed\sensors.yaml

2.2.5.2 Raw IMU data save location

project_path\vSLAMCache\run\ReplayCamera0output\imu\Data.csv

2.2.5.3 INS data saved in Voyis Format

project_path\vSLAMCache\run\ReplayCamera0output\geo\Data.csv

2.3 How to export the point cloud

Point clouds produced in a coordinate system (georeferenced point cloud) with INS aiding are not compatible with EIVA NaviSuite Software if they are exported directly from Discovery Powered by NaviSuite. To make them compatible with the rest of NaviSuite products (specifically with NaviModel for further post processing) users need to export the Point cloud from Discovery Powered by NaviSuite by right clicking on the point cloud in **Project Tree** and selecting **Create Georeferenced Pointcloud**.

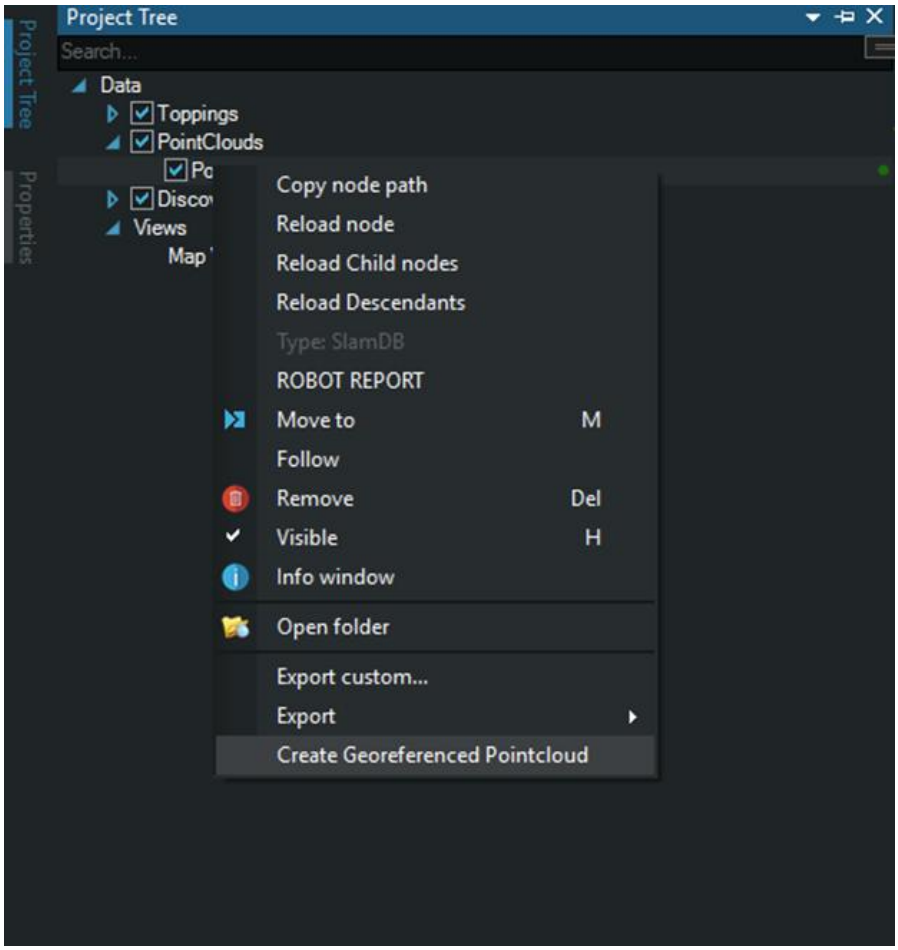

Figure 23 Exporting georeferenced point cloud

2.3.1 XYZ export

To export a structured point cloud, right-click on the point cloud and select **Export** > **As .xyz**.

To export an unstructured point cloud, right-click on the point cloud and select **Export** > **Points**.

2.3.2 CSV export

To export a structured point cloud, right-click on the point cloud and select **Export** > **As .csv**.

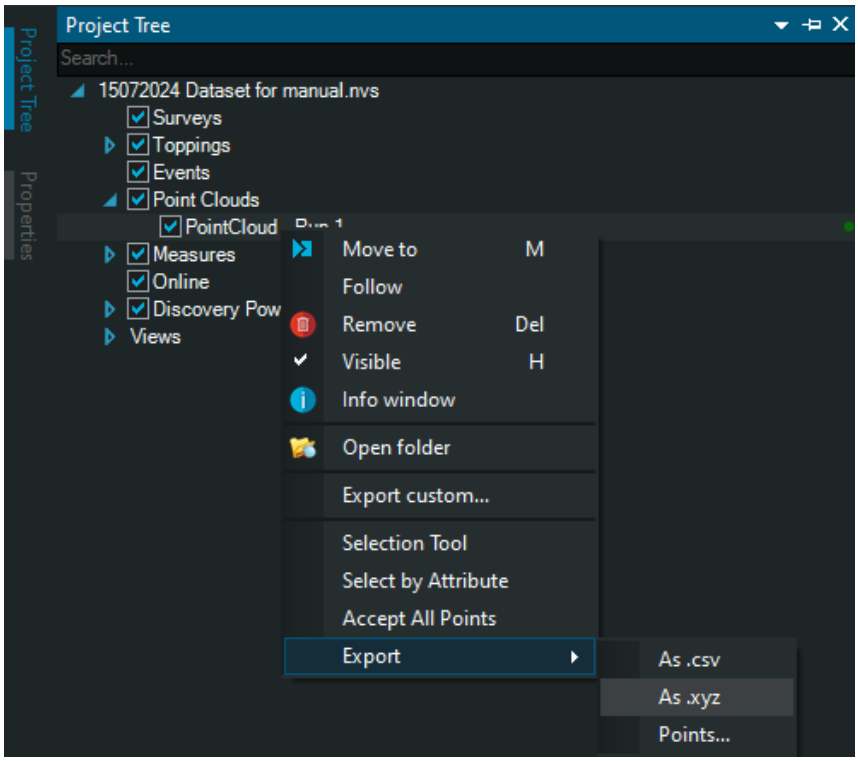

Figure 24 Discovery Powered by NaviSuite – Export as .xyz

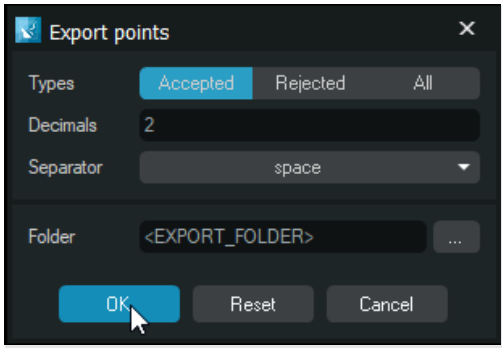

Figure 25 Discovery Powered by NaviSuite – Export points

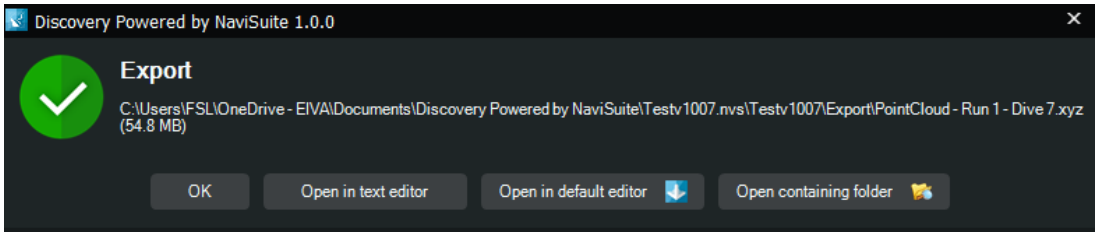

Figure 26 Discovery Powered by NaviSuite – Export location

2.3.3 LAS and LAZ export

To export point cloud in LAS or LAZ format, right-click on the point cloud and select **Export custom**

Under the submenu there are different options that can be used for export.

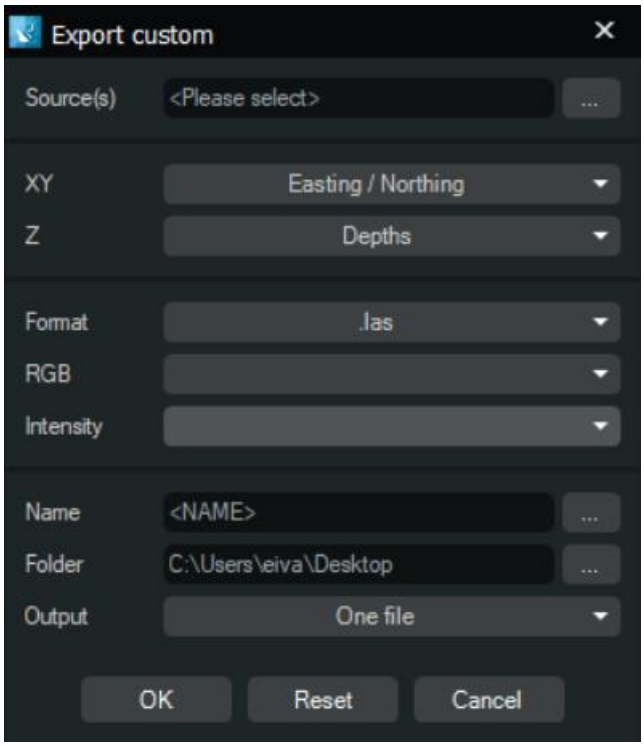

Figure 27 Discovery Powered by NaviSuite – Default menu for export of .LAS format

2.4 Storage indicator

2.4.1 Remaining storage check on startup

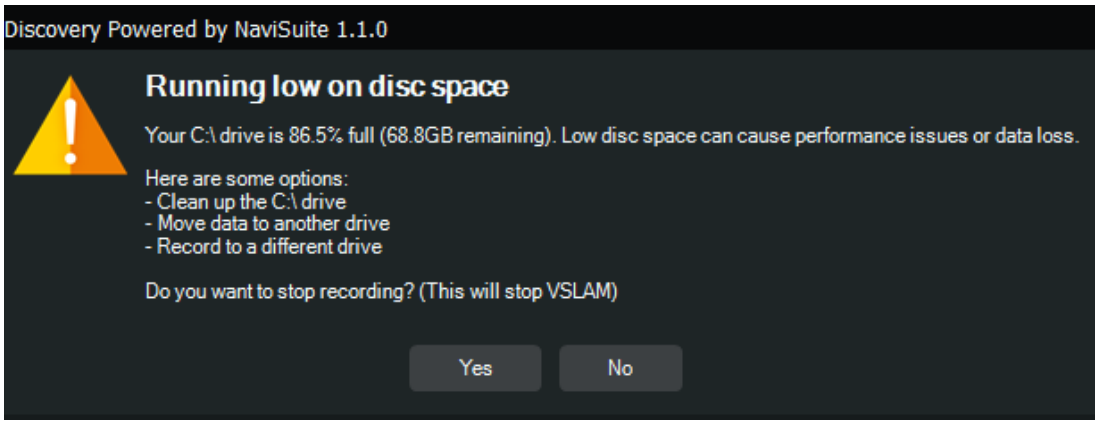

Figure 28 Discovery Powered by NaviSuite – remaining storage check pop up

On the startup of the Discovery Powered by NaviSuite, there is a check performed by the program. The program will check the remaining storage of the drive to which your project is saved every 5 seconds. If there remaining storage is below the threshold (default 100 GB), this warning window will pop-up.

2.4.2 Different states of storage indicator

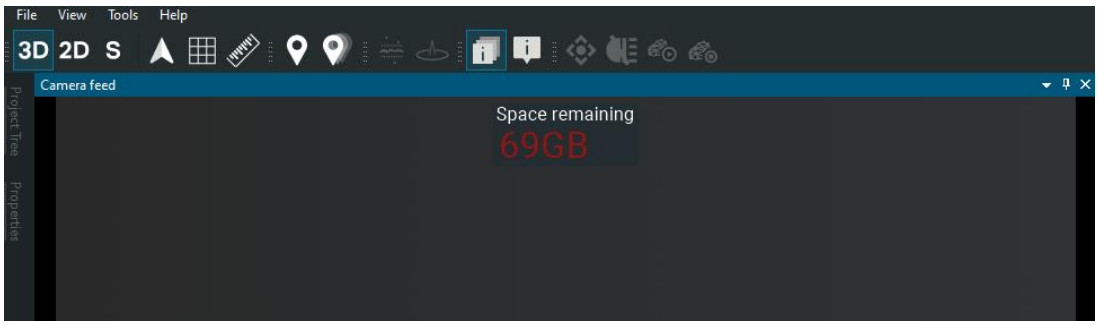

Figure 29 Discovery Powered by NaviSuite – Space remaining in GB

The storage indicator is not shown by default. It shows up automatically once the red warning state is entered (and disappears again automatically, if the red state is exited). The red warning state shows limited space remaining on the storage device.

The user can choose to show it themselves by creating a dashboard or similar.

Note: If it has been deleted, it can be found in live data using any of the live data displays (**Data view**, **Dashboard**)

By default, Discovery Powered by NaviSuite displays the red warning with *Space remaining* in GB (gigabytes) when the storage drive has less than 100 GB of space remaining. When the indicator is showing in a dashboard or text overlay, the storage statuses that can be seen are as follows:

- Green – When you have more than 150 GB remaining on the drive where your project is located.
- Yellow – When you have between 101GB and 150GB remaining.
- Red – When you have 100GB or less space remaining on the drive, along with the window appearing shown in Figure 28 Discovery Powered by NaviSuite – remaining storage check pop up.

When the warning state occurs, you can choose **Yes** or **No**:

- **Yes** – Stops all recording. If you choose to stop recording, NaviSuite VSLAM will stop.
- **No** – Do nothing and continue as before. If you choose to do nothing, Discovery Powered by NaviSuite will open or continue as usual.

Regardless of the option you choose the pop-up should not show up again, unless you go 5GB above the threshold (ie 105GB by default or more).

All space indicators (values in live data) reflect the drive your current project is saved to, so if you save using **Save As** to a different drive (including an external drive), it will update within 5s to reflect space remaining on the newly selected drive.

2.5 Map view layout

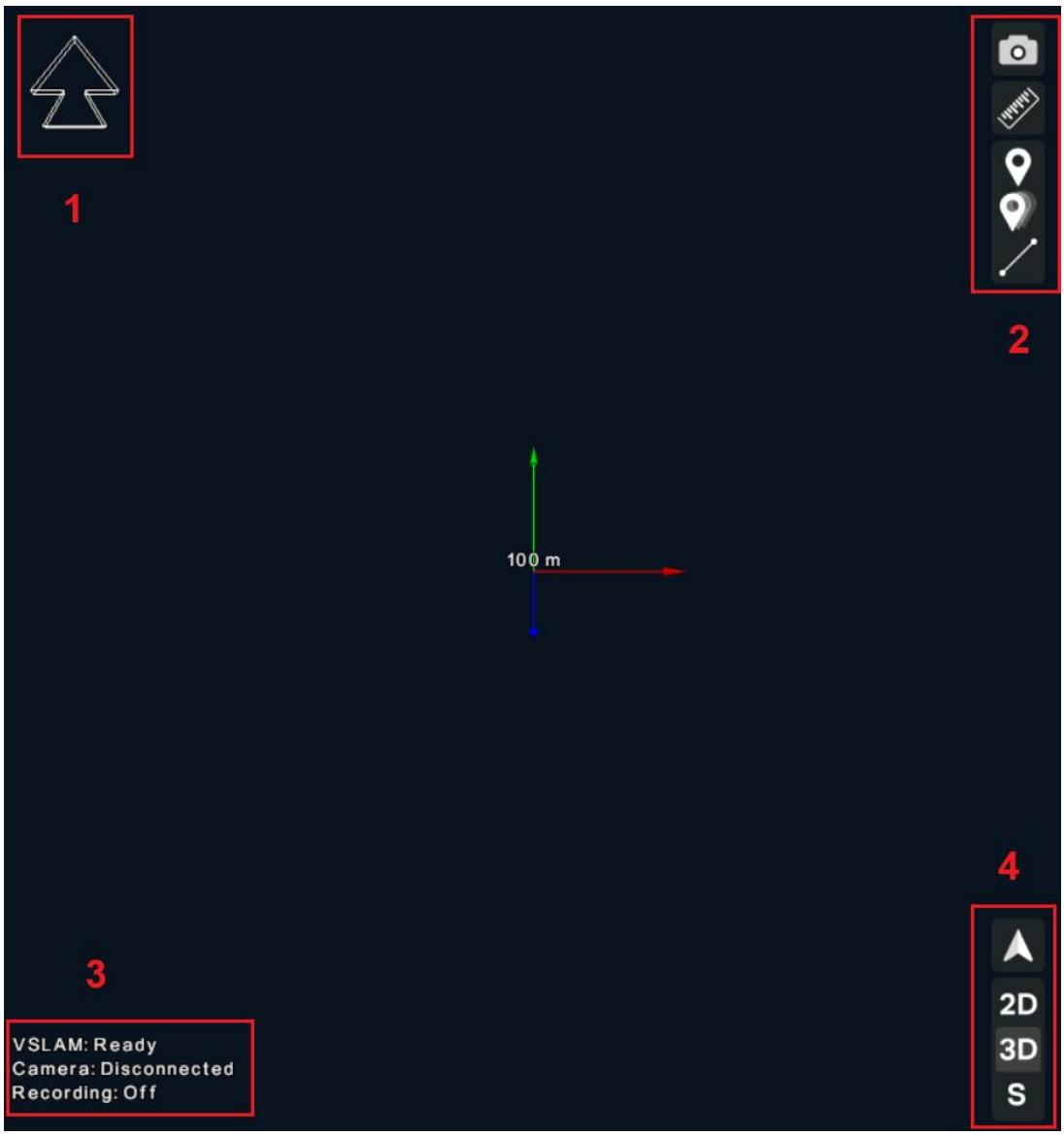

Figure 30 Discovery Powered by NaviSuite – Map view

In Discovery Powered by NaviSuite version 1.1, as default, EIVA has introduced a new **Map View** layout. The new **Map View** layout has 3 toolbars and 1 information field.

In the top left corner (1) of the **Map View** there is the **North arrow** that has multiple applications:

- Pressing once will align the view with the north up

- Pressing a second time will start scrolling between 3D and 2D view

In the top right corner (2) of the **Map View** there is a toolbar with multiple functions:

- Screenshot function
- Measure function
- New Waypoint
- New Waypoints
- New digitised line

In the bottom left corner (3) of the **Map View** there is an information field showing statuses:

- VSLAM status
- Camera status
- Recording status

In the bottom right corner (4) of the **Map View** there is a toolbar that has multiple view functions:

- Align view to north up
- 2D View
- 3D View
- Side View

2.5.1 VSLAM, Loop closure indicator, Camera and Recording status shown in Map View

The following statuses can be seen in the information field, (3) in figure 25.

VSLAM:

- *Go to last tracked VSLAM* – Status will appear if VSLAM has lost track
- *Tracking OK* – VSLAM is running under normal circumstances
- *Align camera with structure* – VSLAM cannot match the features in the image with any of the features in the map
- *No frames received* – VSLAM is not receiving any images from the camera
- *Internal error* – Communication between Discovery Powered by NaviSuite and NaviSuite VSLAM is not working
- *Ready* – VSLAM is not started and is in ready status

Camera:

- *Connected* – Discovery Powered by NaviSuite is receiving messages from the camera

- *Disconnected* – Discovery Powered by NaviSuite is not receiving messages from the camera
- *Awaiting Camera* – Discovery Powered by NaviSuite is receiving messages from the camera but no stereo images

Recording:

- **ON** – VSLAM is recording data
- **OFF** – VSLAM is not recording data

Loop closure indicator:

- *Green* – There is no need for a loop closure
- *Orange* – Indication that loop closure should be done soon to keep consistency of the data
- *Red* – User should consider doing the loop closure if possible.

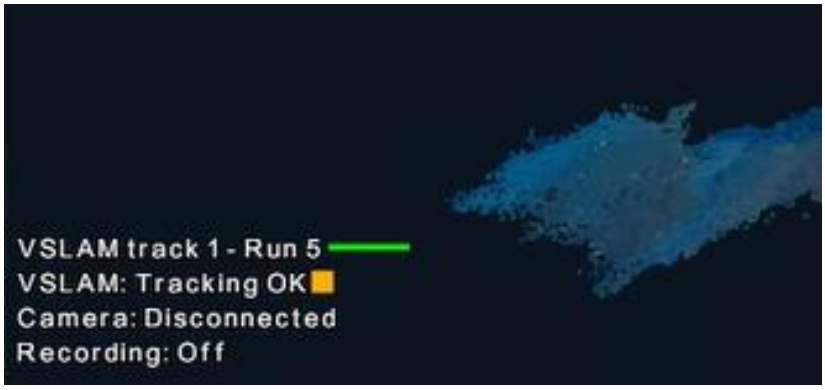

Figure 31 Discovery Powered by NaviSuite – Loop closure indicator light shown next to Tracking status

3 Useful settings

3.1 Surface grid

Surface grid can be found under **Tools** menu.

The surface grid gives a better perspective for the pilot by helping to see relative distance to the object that is modelled during flight.

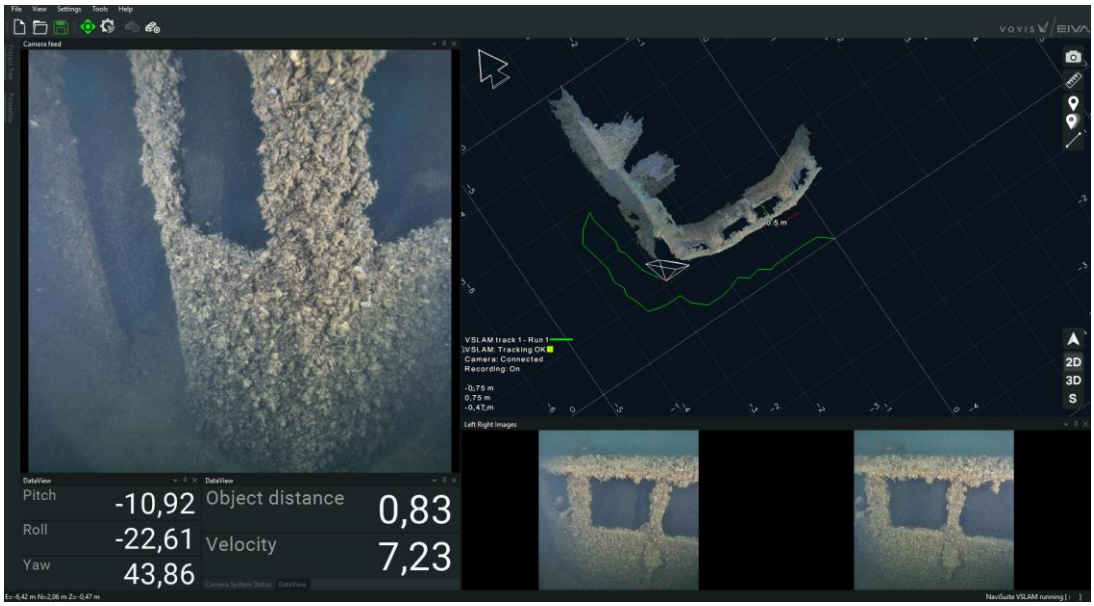

Figure 32 Discovery Powered by NaviSuite with Surface grid turned on

3.2 Measure function

The **Measure** function is enabled by pressing on the ruler icon in the **Map View.**

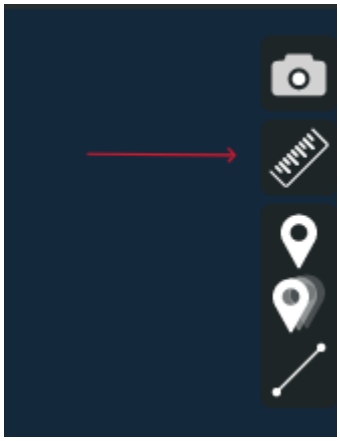

Figure 33 Discovery Powered by NaviSuite – Measure icon placement in the toolbar

Use the **Measure tool** to measure distances in a map view. A new measure object will be created under **Static Objects** in the **Project Tree**. Click a path to get a length at each point, then click **Escape** to end measuring. To remove the measure lines, select the measure object in the **Project Tree** and press the delete key, or delete the object from its context menu.

Measurements are added to the **Project Tree** under **Measures**. Measures can be removed from the project by right clicking the measure and choosing **Remove**.

Measures can be exported to different formats depending in which program you want to continue quality control and processing of the generated point cloud.

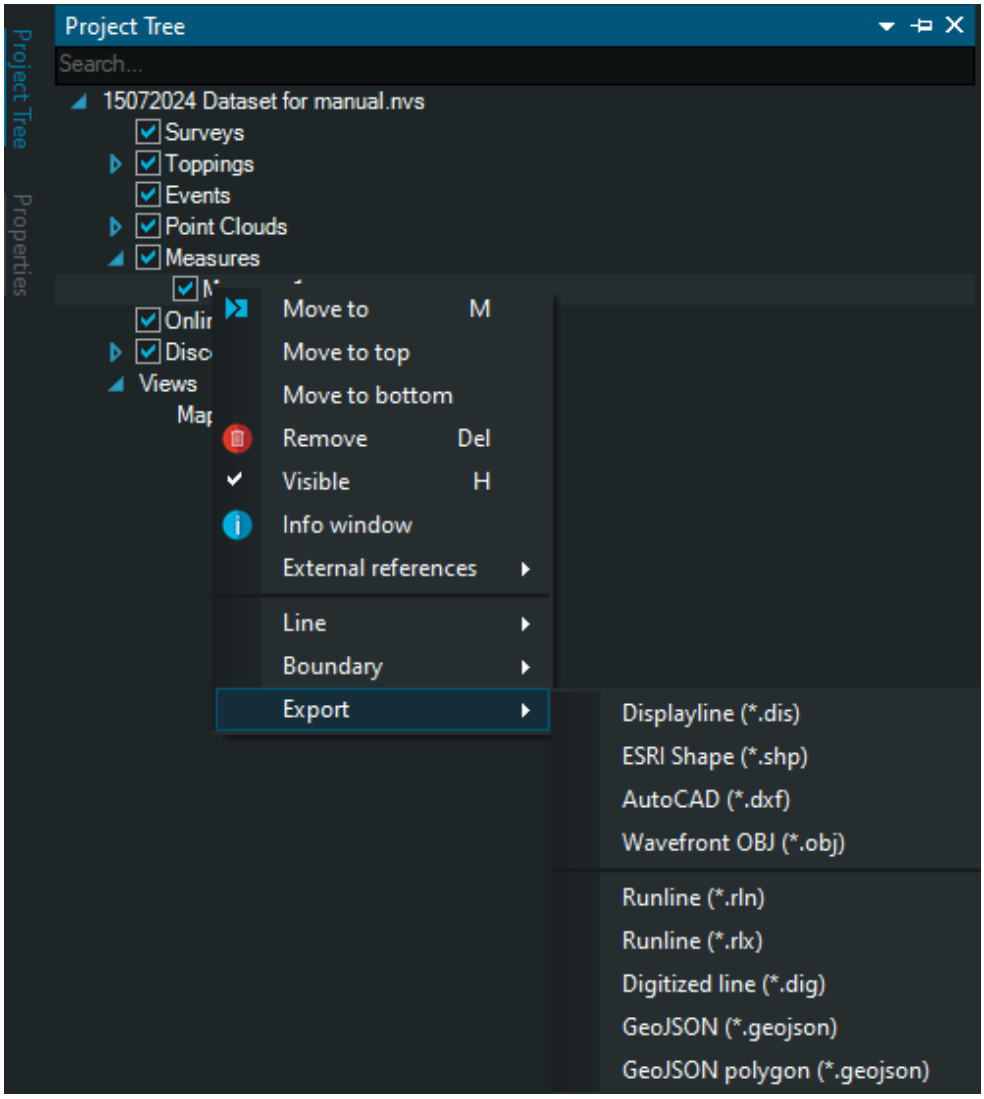

Figure 34 Discovery Powered by NaviSuite – Exporting measurements to different formats

3.3 Point-clouds menu

Access the menu through **Tools** > **Point-clouds**. The menu contains quick access functions to export point clouds to XYZ, the selection tool and a tool for changing the size of the points in a point cloud.

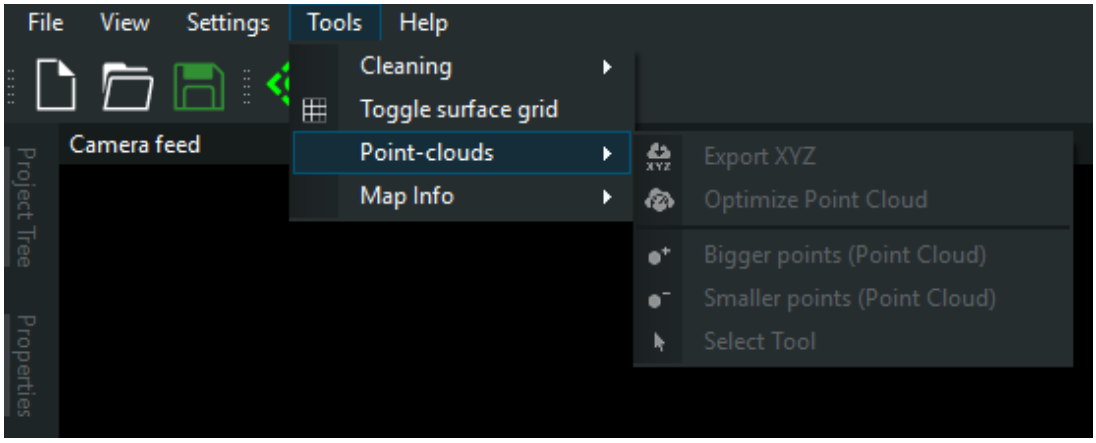

Figure 35 Point-clouds menu

To activate the **Point-clouds menu**, one of the generated point clouds needs to be selected.

The same functions can be found by right clicking on the point cloud in the **Project Tree**. Changing the size of the points in the point cloud can done through the **Properties** pane.

3.4 Waypoint functions

Figure 36 Waypoint and Waypoints icons in Map View

Waypoint functions can be found in the **Map View**.

Please refer to the NaviSuite Helmsman manual on the EIVA Support webpage if you want to know more.

3.5 Log window

Please refer to the NaviSuite Helmsman manual on the EIVA Support webpage if you want to know more.

4 Customising Discovery Powered by NaviSuite UI

4.1 Customising the view(s)

More in-depth information about views can be found in the NaviSuite Helmsman manual on the EIVA Support webpage. Here are some quick tips for getting started.

The different views and windows can be tailored to fit your needs. The views can be either floating or docked. Docking a view is done by dragging the view until the icon of a rectangle with and arrow in front appears. Leaving the view in the screen without using the docking means the view is floating.

Figure 37 Docking button

Left-click the header of any view you want to move, and the docking icon will appear at possible docking destinations on the display. Then, drag and dock it to the desired icon. The window will then be docked.

If you want to float the view, you can left-click the header of the docked view and drag and drop it anywhere where the docking icon is not visible.

Here are two possible view layouts when using Discovery Powered by NaviSuite, one for an ROV pilot and one for a surveyor.

4.1.1 Example view layout for an ROV pilot

Figure 38 Suggested layout for ROV pilots – Camera feed (left), Data view - Object distance and RPH values (bottom left), Map View (centre), Camera and lights control (top right), Camera System Status (centre right), Last tracked keyframe (bottom right)

4.1.2 Example view layout for a surveyor

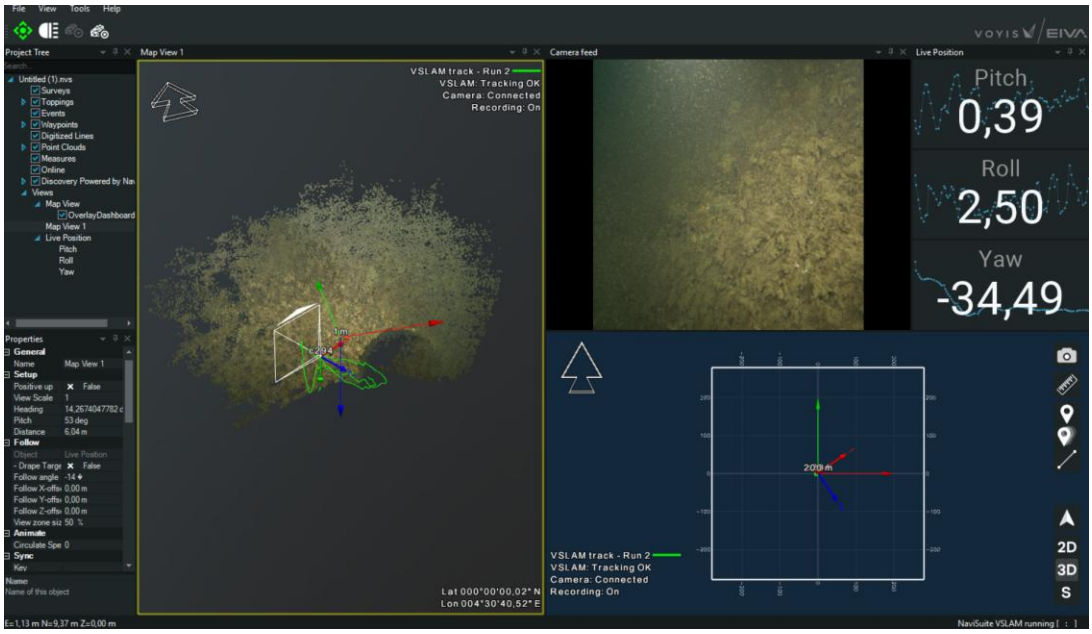

Figure 39 Suggested layout for Surveyors – (left to right) Project Tree (top), Properties(bottom), Map View 1, Camera feed (top left), Dashboard view (top right), Map View (bottom)

4.2 Video view

Please refer to NaviSuite Mobula Manual on EIVA's Support webpage if you want to know more.

4.3 Overlays

Please refer to NaviSuite Mobula Video overlay in the FreshDesk FAQ.

There is also an FAQ section on how to set text overlay in FreshDesk.

4.3.1 Pilot Overlay

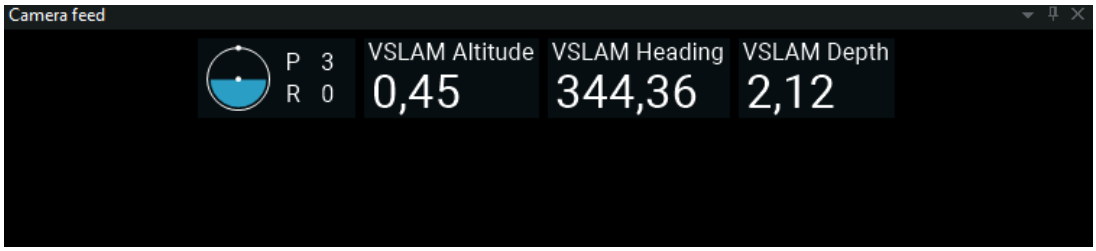

Figure 40 Camera System status

The **Pilot Overlay** is activated through **Camera Control** window. The **Pilot Overlay** will overlay data needed for the ROV pilot over the **Camera Feed** window.

If the user has configured INS to output data to the Discovery camera through the **Sensor Settings** window, then the values shown in the **Pilot Overlay** are the INS sensor values. This overlay will ease piloting the ROV as it gives all the relevant data to the pilot.

If some of the data is not needed, then the data can be unticked in the **Project Tree**.

4.4 Dashboards

In Discovery Powered by NaviSuite, there are two specific data dashboards that can be added to the view:

- **VSLAM status** – separated into Recording status and VSLAM Status
- **VSLAM Data** – shows attitude of the camera

Please refer to this NaviSuite Helmsman manual on the EIVA Support webpage if you want to know more.

4.5 Graph view

In Discovery Powered by NaviSuite, there is a specific graph that can be added to the view:

- **VSLAM Data** – shows the attitude of the camera

Please refer to this NaviSuite Helmsman manual on the EIVA Support webpage if you want to know more.

4.6 Toggle left right images

Figure 41 Left right images view

Clicking **Toggle left right images** will open a separate window that displays a synchronized preview of the left and right stereo camera feed.

The update rate is the same as the chosen **Data Publishing Rate (fps)**.

These images are one-quarter (0.25x) the resolution of the normal still images, allowing them to be streamed even when bandwidth is limited.

Voyis and EIVA recommend using the preview stills images for robotics applications where the full resolution images may be too large to stream over network connections.

4.7 Toggle camera system status

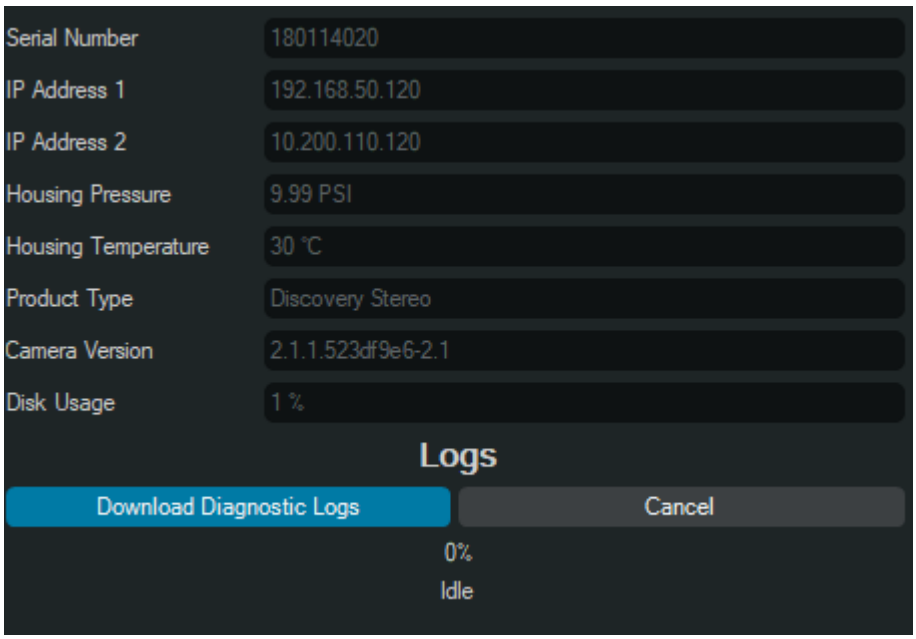

Figure 42 Camera System status

- **Serial Number** – Discovery camera product identifier
- **IP Address** – IP address associated with the Discovery product
- **Housing Pressure** – Internal housing pressure, held under vacuum at approximately 10 PSI.
    - **Note**: Verify before deployment, as a drop can indicate a housing leak.
- **Housing Temperature** – Temperature of the internal control electronics.
- **Product Type** – indicates if Discovery Stereo (VSLAM and piloting) or Discovery Mono (only piloting) camera is connected.
- **Camera Version** – Software version, useful for reporting to EIVA Helpdesk / Voyis Support.
- **Disk Usage** (%) – Percent of onboard storage used.
- **Download Diagnostic Logs –** Selecting this will download the onboard logs which can be used for debugging. In case there are issues with the camera these logs need to be sent to Voyis technical support.